Check for updates

ADVANCED HEALTHCARE MATERIALS
www.advhealthmat.de

Injectable Double Crosslinked Hydrogel-Polypropylene Composite Mesh for Repairing Full-Thickness Abdominal Wall Defects

Yixin Zhao, Xiaopei Li, Ni Sun, Yan Mao, Teng Ma, Xiangping Liu, Tao Cheng, Xiangyu Shao, Haifeng Zhang, Xianggang Huang, Junsheng Li, Ningping Huang,* and Haibo Wang**

Abdominal wall defects are common clinical diseases, and mesh repair is the standard treatment method. The most commonly used polypropylene (PP) mesh in clinical practice has the advantages of good mechanical properties, stable performance, and effective tissue integration effect. However, direct contact between abdominal viscera and PP mesh can lead to severe abdominal adhesions. To prevent this, the development of a hydrogel-PP composite mesh with anti-adhesive properties may be an effective measure. Herein, biofunctional hydrogel loaded with rosmarinic acid is developed by modifying chitosan and Pluronic F127, which possesses suitable physical and chemical properties and commendable in vitro biocompatibility. In the repair of full-thickness abdominal wall defects in rats, hydrogels are injected onto the surface of PP mesh and applied to intraperitoneal repair. The results indicate that the use of hydrogel-PP composite mesh can alleviate abdominal adhesions resulting from traditional PP mesh implantation by decreasing local inflammatory response, reducing oxidative stress, and regulating the fibrinolytic system. Combined with the tissue integration ability of PP mesh, hydrogel-PP composite mesh has great potential for repairing full-thickness abdominal wall defects.

1. Introduction

Abdominal wall defects are common surgical conditions, usually caused by factors such as surgery, trauma, or infection. Over the past few decades, tension-free repair has been considered the standard treatment for abdominal wall defects, and mesh materials have been widely applied to cover and reinforce the weak tissues of the abdominal wall.[1,2] At present, polypropylene (PP) mesh is the main prosthetic material in the field of hernia repair. PP mesh not only has the characteristics of relatively low price and excellent mechanical properties but also can stimulate the proliferation of fibrous tissue and effectively integrate with the host tissue, so as to realize the repair and reconstruction of abdominal wall defects.[3,4] However, when PP mesh comes into direct contact with abdominal viscera, it will cause abdominal adhesions, leading to long-term complications such as chronic abdominal pain, intestinal obstruction, and female infertility. Consequently, the surgeon may need to perform reoperations, adding to treatment expenses. Given this, it has become increasingly imperative to prevent or lessen postoperative adhesions. With the continuous development of materials science and tissue engineering, numerous anti-adhesion meshes have been developed for intraperitoneal repair, but there is still no ideal mesh that can meet a variety of different

Y. Zhao, Y. Mao, T. Ma, X. Liu, H. Wang
Breast Disease Center
The Affiliated Hospital of Qingdao University
Qingdao 266071, China
E-mail: hbwang66@qdu.edu.cn

Y. Zhao, H. Zhang, X. Huang, J. Li
School of Medicine
Southeast University
Nanjing 210009, China
E-mail: 101008114@seu.edu.cn

X. Li
Department of Cardiovascular Surgery
The First Affiliated Hospital of Nanjing Medical University
Nanjing 210029, China

N. Sun
Department of Radiation Oncology
Jinling Hospital
Medical School of Nanjing University
Nanjing 210002, China

T. Cheng, X. Shao, J. Li
Department of General Surgery
The Affiliated Zhongda Hospital of Southeast University
Nanjing 210009, China

N. Huang
School of Biological Science and Medical Engineering
Southeast University
Nanjing 210096, China
E-mail: nphuang@seu.edu.cn

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/adhm.202304489

DOI: 10.1002/adhm.202304489

 © 2024 Wiley-VCH GmbH

clinical needs at the same time. Previous studies have shown that adding a physical barrier to the surface of PP mesh appears to be a feasible measure, which can combine the advantages of PP mesh in promoting abdominal wall integration, while isolating PP mesh from the viscera to reduce the occurrence of adhesions, thereby achieving efficient repair of abdominal wall defects.[5,6]

Hydrogel, a polymer with a 3D network structure, is widely used in the field of tissue engineering, wound dressing, and drug delivery due to its high hydrophilicity, good biocompatibility, and porous structure similar to extracellular matrix.[7–9] Previous studies have reported that hydrogels can prevent peritoneal adhesions by isolating the injured sites from abdominal viscera, inhibiting local inflammatory responses, and reducing abnormal collagen deposition.[10–12] The construction of a hydrogel-PP composite mesh with anti-adhesion properties may be a promising approach to promote the intraperitoneal repair of abdominal wall defects and prevent the occurrence of abdominal adhesions. In recent years, injectable hydrogels and self-healing hydrogels have attracted attention due to their characteristics of easy manipulation, good shape adaptability, and rapid self-healing. The preparation of injectable and self-healing hydrogels can be achieved through a variety of physical or chemical crosslinking reactions. Nonetheless, hydrogels prepared from a single crosslinking network may have disadvantages such as relatively poor mechanical properties and unstable structures.[13,14] Therefore, enhancing the stability of hydrogels while maintaining their injectability and self-healing properties remains a challenge. As a physical barrier between the abdominal viscera and PP mesh, it is of great significance to design a hydrogel with ease of use, stable structure, good biocompatibility, and excellent anti-adhesion ability for the prevention of abdominal adhesions in the intraperitoneal repair of abdominal wall defects.

Chitosan (CS), a natural polymer widely found in nature, is produced by the deacetylation of chitin and has been widely used in the field of biomedical materials due to its good antibacterial activity, biocompatibility, biodegradability, and non-antigenicity. The main disadvantage of chitosan is its poor solubility, but its surface features numerous active groups, such as amino groups and hydroxyl groups, which can be modified to introduce new groups and enhance its water solubility.[15–17] Pluronic F127, a typical amphiphilic triblock copolymer of polyoxyethylene-polyoxypropylene-polyoxyethylene (PEO-PPO-PEO), is a common thermosensitive nonionic surfactant. Pluronic F127 polymer molecules self-assemble into micelles with water-insoluble PPO as the core and water-soluble PEO as the shell, so that drugs with poor water solubility can be encapsulated into the hydrophobic PPO core.[18,19] As a natural phenolic acid compound, rosmarinic acid (RA) has a variety of immunomodulatory functions, including anti-inflammatory, antioxidative, and anti-tumor properties, which has attracted increasing attention from researchers.[20,21] However, the role of rosmarinic acid in repairing abdominal wall defects and preventing abdominal adhesions is still poorly understood. In addition, the poor water solubility and low bioavailability limit its clinical application.[22,23]

In this work, we synthesized methacrylic anhydride modified chitosan (CS-MA) and benzaldehyde-terminated Pluronic F127 (F127-CHO), and then loaded rosmarinic acid into F127-CHO (F127-CHO@RA). Based on Schiff base reaction and photopolymerization reaction, double crosslinked hydrogels with injectable and self-healing properties were prepared (**Scheme 1a**). The physicochemical and biological properties of the hydrogels were subsequently evaluated. Finally, we established a full-thickness abdominal wall defects model in rats, and the hydrogels were injected in situ onto the surface of PP mesh for intraperitoneal repair, thus effectively creating isolation between the abdominal viscera and PP mesh (Scheme 1b). We evaluated whether the hydrogel-PP composite mesh could repair abdominal wall defects while reducing abdominal adhesions, and its mechanism was further investigated.

2. Results and Discussion

2.1. Successful Synthesis of PCMFC and PCMFC@RA Hydrogels

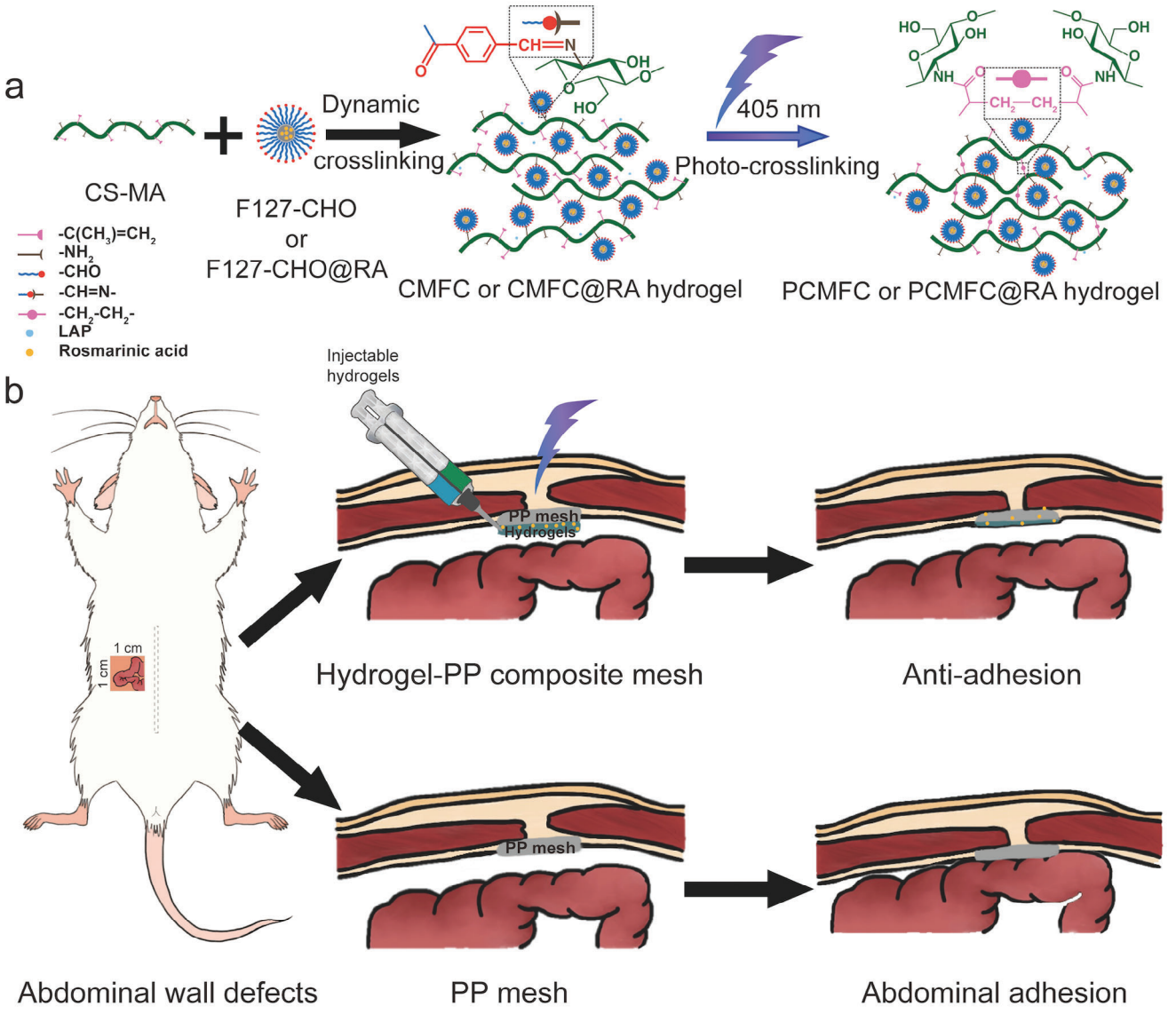
2.1.1. Preparation and Characterization

In order to impart CS with water solubility and photoresponsiveness, chitosan methacryloyl (CS-MA) was synthesized by grafting CS with methacrylic anhydride (MA) through a one-step chemoselective N-acylation reaction (Figure S1a, Supporting Information). As shown in Figure S1b (Supporting Information), the new peak at 840 cm^{-1} in the Fourier transform infrared (FTIR) spectrum of CS-MA corresponded to the out-of-plane deformation vibration peak of C—H on the C=C double bond, indicating the successful grafting of methacryloyl groups onto chitosan. Due to the formation of new amide bonds, the characteristic peaks observed at 1649, 1527, and 1323 cm^{-1} corresponded to the amide I band (C=O stretching), amide II band (N—H deformation), and amide III band (C—N stretching), respectively. In addition, the characteristic absorption peak (1730–1740 cm^{-1}) of the ester group was not detected, indicating that methacrylic anhydride was not combined with the hydroxyl group. Therefore, the chemoselective N-acylation reaction of MA with the primary amino groups of CS occurred. The chemical structure of CS-MA was further confirmed with ^{1}H nuclear magnetic resonance (^{1}H NMR) spectrum. Two signal peaks with chemical shifts of 5.32 and 5.63 ppm due to vinyl protons were visible. Moreover, methyl protons of methacrylic anhydride residues at 1.84 ppm were clearly observed (Figure S1c, Supporting Information). Next, F127-CHO was synthesized by grafting 4-formylbenzoic acid onto the terminal of Pluronic F127 by esterification (Figure S2a, Supporting Information). To confirm the introduction of aldehyde groups, FTIR and ^{1}H NMR analyses were performed. In contrast to Pluronic F127, the characteristic peak at 1725 cm^{-1} in the FTIR spectrum of F127-CHO was attributed to the C=O stretching vibration, indicating the successful introduction of aldehyde groups (Figure S2b, Supporting Information). In addition, the ^{1}H NMR spectrum of F127-CHO clearly showed new resonance signals at 10.10, 8.20, and 7.96 ppm attributed to aldehyde groups and aromatic ring protons, which further confirmed the successful synthesis of F127-CHO (Figure S2c, Supporting Information).

The Schiff base reaction between the amino groups in CS-MA and the aldehyde groups in F127-CHO formed the initial dynamic crosslinked network of the hydrogel, named CMFC hydrogel. Next, the photoinitiator lithium phenyl-2,4,6-trimethylbenzoylphosphinate (LAP) in the hydrogel attacked the vinyl in CS-MA under 405 nm blue-violet light, initiating the

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 1. a) Schematic illustration of injectable double crosslinked hydrogel loaded with rosmarinic acid. b) Schematic illustration of the repair of full-thickness abdominal wall defects by PP mesh with or without hydrogels.

photopolymerization as the second network of the hydrogel, and finally forming a dynamic/photo-double crosslinked hydrogel, which was named as PCMFC hydrogel (Scheme 1a).

To explore the characteristics of hydrogels with different component ratios, $PCMFC_X$ hydrogels with different initial concentrations of F127-CHO were then prepared by mixing CS-MA solution and F127-CHO solution in equal volume, where the initial concentration of CS-MA was fixed at 1.5% w/v, and X represented the initial concentration of F127-CHO (i.e., 5%, 10%, and 15% w/v). Gelation time is a meaningful performance indicator of injectable hydrogels in biomedical applications. Measured by the vial inversion method (**Figure 1a**), the gelation time of the Schiff base reaction of hydrogels with different component ratios ranged from 41.5 to 141.9 s (Figure S3a, Supporting Information). With the increase in the concentration of F127-CHO, more aldehyde groups reacted with amino groups, and the gelation speed gradually increased. If the gelation time is too long, the hydrogel in a fluid state cannot adhere firmly to the injured tissue. However, fast gelation may fail to attain a homogeneous network structure, making it difficult for surgeons to handle the hydrogel.[24] $PCMFC_{10}$ hydrogel exhibited a gelation time of 64.5 s, which may have better clinical practicability.

2.1.2. Microstructure, Swelling, and Degradation Abilities

Scanning electron microscopy (SEM) images showed that PCMFC hydrogels exhibited a porous and interconnected network structure resembling extracellular matrices, enhancing their potential utility as tissue remodeling materials (Figure 1b). As the F127-CHO content increased, the average pore sizes of $PCMFC_5$, $PCMFC_{10}$, and $PCMFC_{15}$ hydrogels decreased from

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

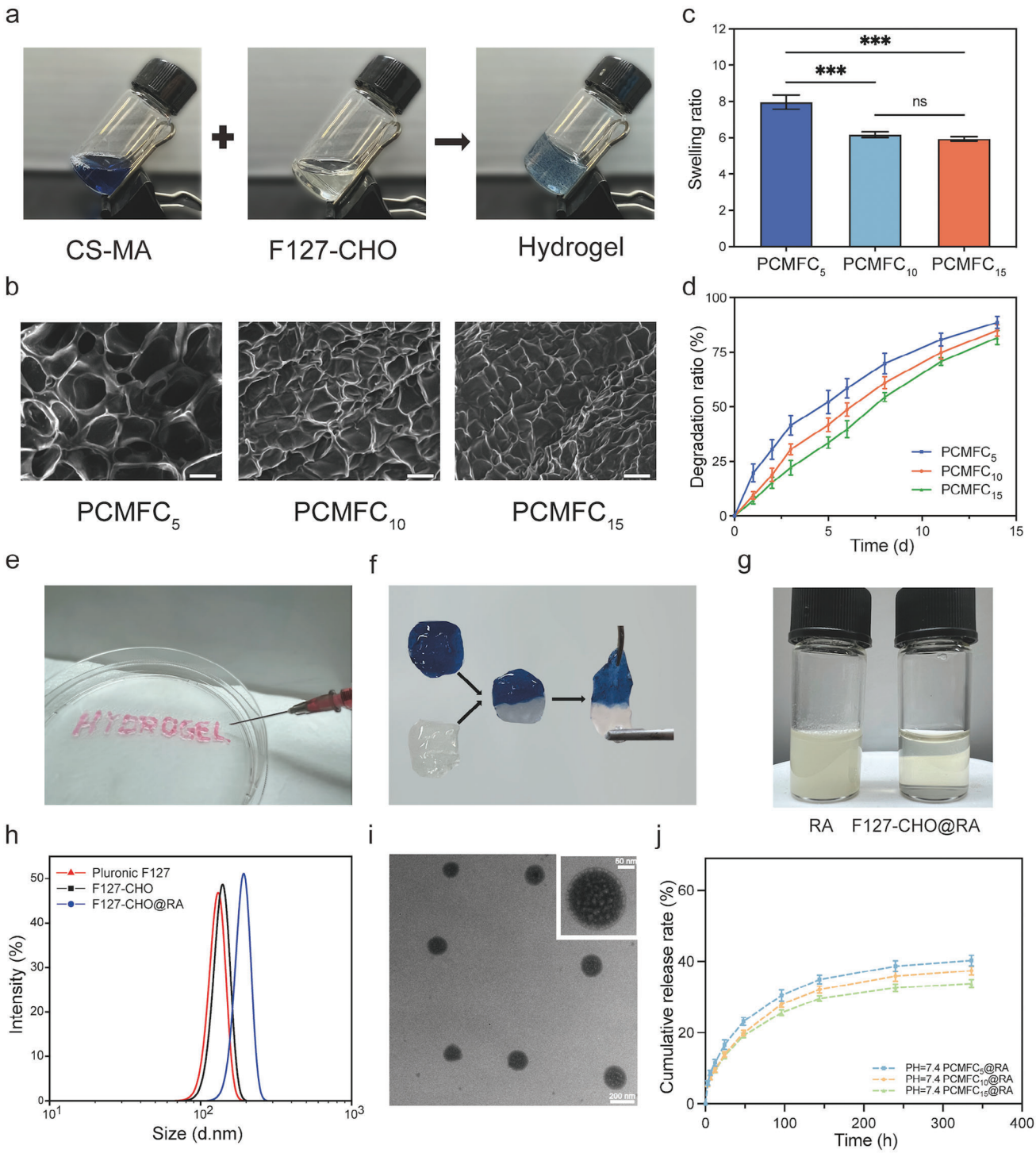


Figure 1. Physical and chemical properties of hydrogels. a) Gel formation process of PCMFC hydrogels. b) SEM images of PCMFC hydrogels. Scale bar: 100 μm. c) Swelling ratios of PCMFC hydrogels in PBS. d) Degradation ratio of PCMFC hydrogels in PBS with lysozyme. e) Injectability of preformed $PCMFC_{10}$ hydrogel. f) Macroscopic self-healing performance of $PCMFC_{10}$ hydrogel. g) Photograph of F127-CHO and F127-CHO@RA in water. h) Particle sizes of Pluronic F127, F127-CHO, and F127-CHO@RA micelles detected by DLS. i) TEM images of F127-CHO@RA micelles. Scale bar: 200 nm. Inset: Scale bar: 50 nm. j) In vitro release kinetics of rosmarinic acid in PCMFC@RA hydrogels in PBS at pH 7.4. All data are presented as mean ± SD ($n = 3$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

158.6, 86.6 to 72.1 μm, respectively, indicating that the crosslinking density of the hydrogels increased (Figure S3b, Supporting Information).

The proper swelling properties of hydrogels are beneficial to promote tissue repair and regeneration by absorbing tissue surplus exudates and blood. The swelling ratio of $PCMFC_5$, $PCMFC_{10}$, and $PCMFC_{15}$ hydrogels decreased from 7.96, 6.17, and 5.94, respectively, attributing to the increase of crosslinking density (Figure 1c). Moreover, the ideal implant material for abdominal wall defect repair should have appropriate degradation properties, which can not only provide a complete physical barrier before the tissue recovers entirely to prevent abdominal adhesions but also avoid chronic inflammation caused by excessively slow degradation. Generally, peritoneal mesothelial cells will heal within 7–10 d after injury. During this period, it is recommended to minimize contact between the damaged peritoneum and abdominal organs to decrease the formation of adhesions.[25] The results showed that after incubation in phosphate-buffered saline (PBS) with lysozyme at 37 °C for 14 d, the degradation rates of $PCMFC_5$, $PCMFC_{10}$, and $PCMFC_{15}$ hydrogels were 88.7%, 85.0%, and 81.7%, respectively, indicating that the three groups of hydrogels had good biodegradability (Figure 1d).

2.1.3. Rheological Properties, Injectable, and Self-Healing Performance

We hypothesize that due to the introduction of photo-crosslinking network, the mechanical properties of double crosslinked hydrogels will be higher than those of single Schiff base crosslinked hydrogels. To test this hypothesis, we prepared hydrogels with and without photo-crosslinking and then compared their mechanical properties. The storage modulus (G') and loss modulus (G'') of the hydrogels versus time were detected at constant strain and frequency to obtain the rheological properties of different hydrogels. The G' and G'' of the hydrogels remained stable, with G' being consistently higher than G'', indicating the elastic behavior and gel state of the hydrogels. For single crosslinked CMFC hydrogels (Figure S4a, Supporting Information), with the increase of F127-CHO content, G' increased by over an order of magnitude from 240.1 to 3015.5 and 5177.2 Pa, respectively, suggesting that the mechanical properties of hydrogels increased with the degree of crosslinking. For double crosslinked PCMFC hydrogels (Figure S4b, Supporting Information), with the increase of F127-CHO content, G' also increased, and the G' of $PCMFC_{15}$ hydrogel could reach 6639.8 Pa. In addition, due to the free radical photopolymerization between the C═C bonds of CS-MA, the G' of PCMFC hydrogels (Figure S4b, Supporting Information) was further increased compared to that of CMFC hydrogels (Figure S4a, Supporting Information). It shows that there are more crosslinking sites in the double crosslinked hydrogel system, and the introduction of the photo-crosslinking network improves the mechanical properties of the hydrogel. The above results indicate that the mechanical properties of hydrogels can be modulated by the regulation of the double crosslinking structure to meet different biomedical applications.

As shown in Figure 1e, $PCMFC_{10}$ hydrogel could be easily written with the word "HYDROGEL" through a 26 G needle without clogging, indicating that the hydrogel has good injectability and can be injected into the irregular wound of deep tissue. Hence, the defects can be closed by gelation in situ, which has the potential for application in laparoscopic surgery.

External forces such as syringe extrusion and body movement may disrupt the network structure and integrity of the hydrogels, thereby affecting their barrier function. Therefore, the self-healing properties of the hydrogel were the primary mechanical properties of interest in this work, as PP mesh could provide the required mechanical strength for sufficient repair. We evaluated the self-healing properties of the hydrogel through macroscopic and microscopic self-healing tests. In the macroscopic self-healing test (Figure 1f), two pieces of hydrogels (one of which was stained with methylene blue) were cut through the middle, and half of each hydrogel was subsequently spliced together. Within 2 min, the two hydrogels were able to self-heal into a single complete unit due to the rapid reconstruction of imine bonds, and the healed hydrogel could be stretched (Movie S1, Supporting Information), demonstrating the good mechanical properties of the hydrogel. Next, the self-healing ability was further characterized by rheological tests. From the strain amplitude sweep test (Figure S4c, Supporting Information), the G' of $PCMFC_{10}$ hydrogel remained stable at lower strains, which was higher than G'', indicating the elastic behavior of the hydrogel. When the strain exceeded the critical point of 197%, the G' dropped steeply and became lower than G'', suggesting that the hydrogel network collapsed and presented a sol state. From the step-strain sweep test (Figure S4d, Supporting Information), the G' of $PCMFC_{10}$ hydrogel was always higher than G'' at low strain (1%). However, at high strain (500%), G' decreased from 3835.4 to 322.1 Pa, which was lower than G''. However, once the low strain was applied, the G' could recover instantly, which was larger than G'', indicating the transition from sol to gel state and the reconstruction of the hydrogel network. After three cycles, G' still recovered rapidly, proving that the hydrogel had good self-healing properties. Although the G' of the hydrogel had some irreversible loss due to the destruction of the photo-crosslinked network under high strain, the Schiff base reaction rapidly occurred through the contact of free amino groups and aldehyde groups on the fracture surface, and the reversible dynamic covalent crosslinking network guaranteed the superior self-healing performance of the hydrogel.

2.1.4. Drug Loading and Releasing

Pluronic F127 possesses a hydrophobic micellar core, making it appropriate for encapsulating hydrophobic drugs such as rosmarinic acid, a natural polyphenol with excellent anti-inflammatory and antioxidative abilities.[18] F127-CHO loaded with rosmarinic acid, namely, F127-CHO@RA (Figure S5a, Supporting Information), formed a pale yellow transparent solution in water. Compared with the turbid solution formed by pure rosmarinic acid in water, it was shown that F127-CHO greatly enhanced the solubility of rosmarinic acid in water by forming micelles (Figure 1g). The dynamic laser scattering (DLS) analysis revealed the average particle diameters of Pluronic F127, F127-CHO, and F127-CHO@RA micelles to be 130.6, 139.7, and 192.7 nm, respectively, which proved that the size of micelles

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

loaded with rosmarinic acid was larger than that of blank micelles (Figure 1h). Typical transmission electron microscope (TEM) images of F127-CHO@RA micelles are shown in Figure 1i. The micelles were well dispersed and appeared to have a roughly spherical morphology, with an average size of 177.3 nm, which was slightly smaller than that measured by DLS, probably due to dehydration during TEM sample preparation.

The in vitro release behavior of rosmarinic acid in hydrogels (i.e., PCMFC@RA hydrogels) was further investigated in PBS at pH 7.4 and 6.8. As shown in Figure 1j and Figure S5b (Supporting Information), the release of rosmarinic acid showed a fast first and then slow trend. Within 24 h, the release of rosmarinic acid was relatively rapid, which could maintain a high drug concentration for a short time. After the initial burst, the release rate gradually decreased, indicating that the hydrogels had sustained drug release properties and could be used for the long-term release of rosmarinic acid. In addition, the release rate gradually slowed down with increasing F127-CHO content, mainly due to the increased crosslinking density of PCMFC@RA hydrogels. Furthermore, rosmarinic acid in hydrogels exhibited a pH-responsive release behavior, and the release rate increased significantly as the pH value decreased from 7.4 (Figure 1j) to 6.8 (Figure S5b, Supporting Information), which could respond to the evolution of the local microenvironment at different healing stages and achieve efficient tissue repair. This phenomenon was mainly attributed to the pH-sensitive dynamic imine bonds in the hydrogels, and acidic conditions could accelerate the hydrolysis of imine bonds.[26] In addition to this, the amino groups in chitosan were protonated under acidic conditions, and the electrostatic repulsion and hydrophilicity increased,[27] leading to swelling of the hydrogels network, which could also accelerate the release of rosmarinic acid.

2.2. Biological Properties of Hydrogels

2.2.1. In Vitro Anti-Adhesion Properties

The excessive proliferation of fibroblasts will lead to a large amount of extracellular matrix deposition, which eventually promotes the formation of adhesions, so the anti-fibroblast adhesion ability of hydrogels is crucial for preventing postoperative adhesions. NIH-3T3 cells were co-cultured with PCMFC@RA hydrogels or TCPS for 2 d and subjected to LIVE/DEAD staining to assess cell adhesion to the hydrogel surface. As shown in **Figure 2**a, a large number of cells with flat and extended shapes could be observed on the tissue culture plates (TCPS) control. In contrast, there were only a few round cells in hydrogel groups, some growing in clusters and forming multicellular spheres. This phenomenon was attributed to the fact that the hydration layer formed on the highly hydrophilic hydrogels could reduce the adsorption of secreted proteins and weaken cell adhesion.[11]

The anti-nonspecific protein adsorption ability of the hydrogels was determined to further confirm the anti-adhesion properties. After being treated with bovine serum albumin (BSA) solution for 2 h, the protein adsorption amounts of $PCMFC_5$@RA, $PCMFC_{10}$@RA, and $PCMFC_{15}$@RA hydrogel groups were found to be 8.92, 7.73, and 6.75 μg cm^{-2}, respectively, which were significantly less than that of the commercial PP mesh (Figure 2b). The above results indicated that PCMFC@RA hydrogels may be effective in alleviating postoperative adhesions by hindering nonspecific protein adsorption and reducing the interaction between fibroblasts and hydrogels.[28]

2.2.2. In Vitro Hemocompatibility and Cytocompatibility

Good biocompatibility is a prerequisite for the application of biomaterials, and the hemocompatibility of PCMFC@RA hydrogels was evaluated by in vitro hemolysis test. Hemolysis refers to the rupture of erythrocyte membranes, resulting in the release of hemoglobin and other components into the blood, which may be induced by certain drugs or chemicals. The results showed that hydrogels did not cause erythrocyte lysis, evidenced by the almost clear and transparent supernatants (Figure 2c). Meanwhile, compared with positive control group (Triton X-100) and negative control group (normal saline, NS), the hemolysis rates of all hydrogel groups were lower than 1% (Figure 2d), consistent with previous literature.[29] According to the ASTM F756-17 standard of the American Society for Testing and Materials,[30] a hemolysis rate of less than 2% should be considered nonhemolysis and therefore PCMFC@RA hydrogels demonstrated excellent hemocompatibility.

In this study, NIH-3T3 fibroblasts were selected to evaluate the cytocompatibility of PCMFC@RA hydrogels, because fibroblasts play an important role in abdominal wall reconstruction. First, the cell viability of NIH-3T3 cells incubated with PCMFC@RA hydrogels or TCPS for 1, 2, and 3 d was evaluated by the CCK-8 assay. The results (Figure 2e) showed that the cell viability for all hydrogel groups exceeded 85%, proving that the hydrogels exhibited good cytocompatibility. Among them, $PCMFC_{10}$@RA hydrogel group showed the highest cell viability, which could be used as an optional material for abdominal wall defect repair. The cell viability of $PCMFC_{15}$@RA group decreased slightly, probably due to the increased content of aldehyde groups. Therefore, we selected hydrogels with an initial concentration of 10% w/v of F127-CHO (F127-CHO@RA) for subsequent experiments. In addition, the growth status of cells could be visually evaluated by LIVE/DEAD staining, and the fluorescence images (Figure 2f) showed that the cells in each group grew normally with minimal cell death. These results were consistent with those obtained from the CCK-8 assay, demonstrating the good cytocompatibility of PCMFC@RA hydrogels.

2.2.3. In Vitro Antioxidant Capacity

Extensive accumulation of free radicals at the wound site will cause oxidative stress, further leading to lipid peroxidation, DNA damage, and enzyme inactivation, which is not conducive to tissue regeneration. Hydrogels with free radical scavenging ability can promote wound healing by preventing oxidative stress damage.[31,32] The free radical scavenging ability of the hydrogels was evaluated by DPPH· scavenging test, and it was found that the group treated with 5 mg mL^{-1} $PCMFC_{10}$@RA hydrogel could scavenge over 80% of DPPH free radicals. In contrast, $PCMFC_{10}$ hydrogel group without rosmarinic acid showed extremely low DPPH free radical scavenging ability (**Figure 3**a,b).

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

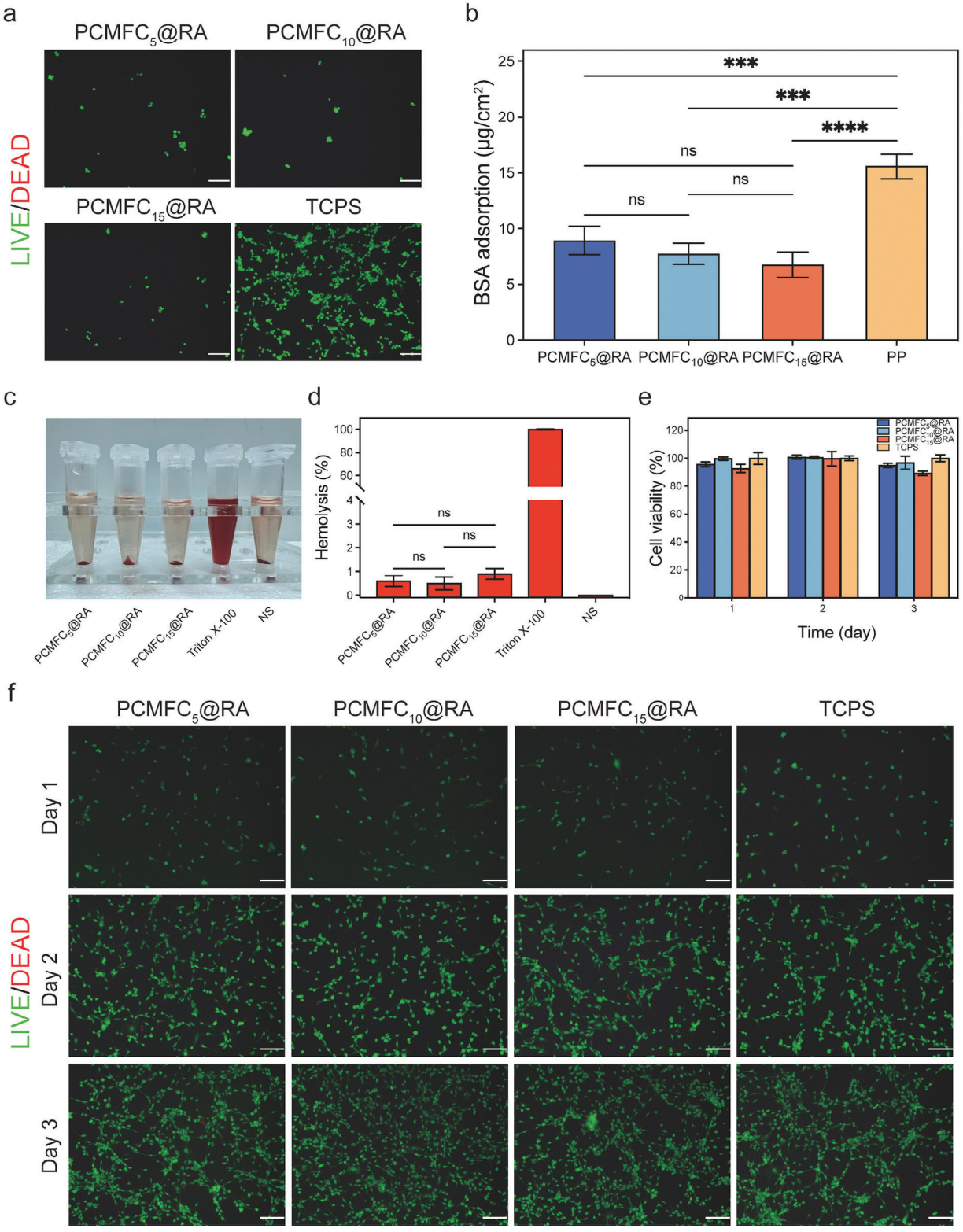


Figure 2. In vitro anti-adhesion properties and biocompatibility of PCMFC@RA hydrogels. a) LIVE/DEAD staining images of NIH-3T3 cells cultured on PCMFC@RA hydrogels and TCPS for 2 d. b) Protein adsorption of PCMFC@RA hydrogels and PP mesh. c) Photograph of hemolysis experiment. d) Hemolysis ratio of PCMFC@RA hydrogels. e,f) Cell viability detected by CCK-8 assay and LIVE/DEAD staining images of NIH-3T3 cells incubated with PCMFC@RA hydrogels and TCPS for 1, 2, and 3 d. Scale bar: 200 μm. All data are presented as mean ± SD ($n = 3$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

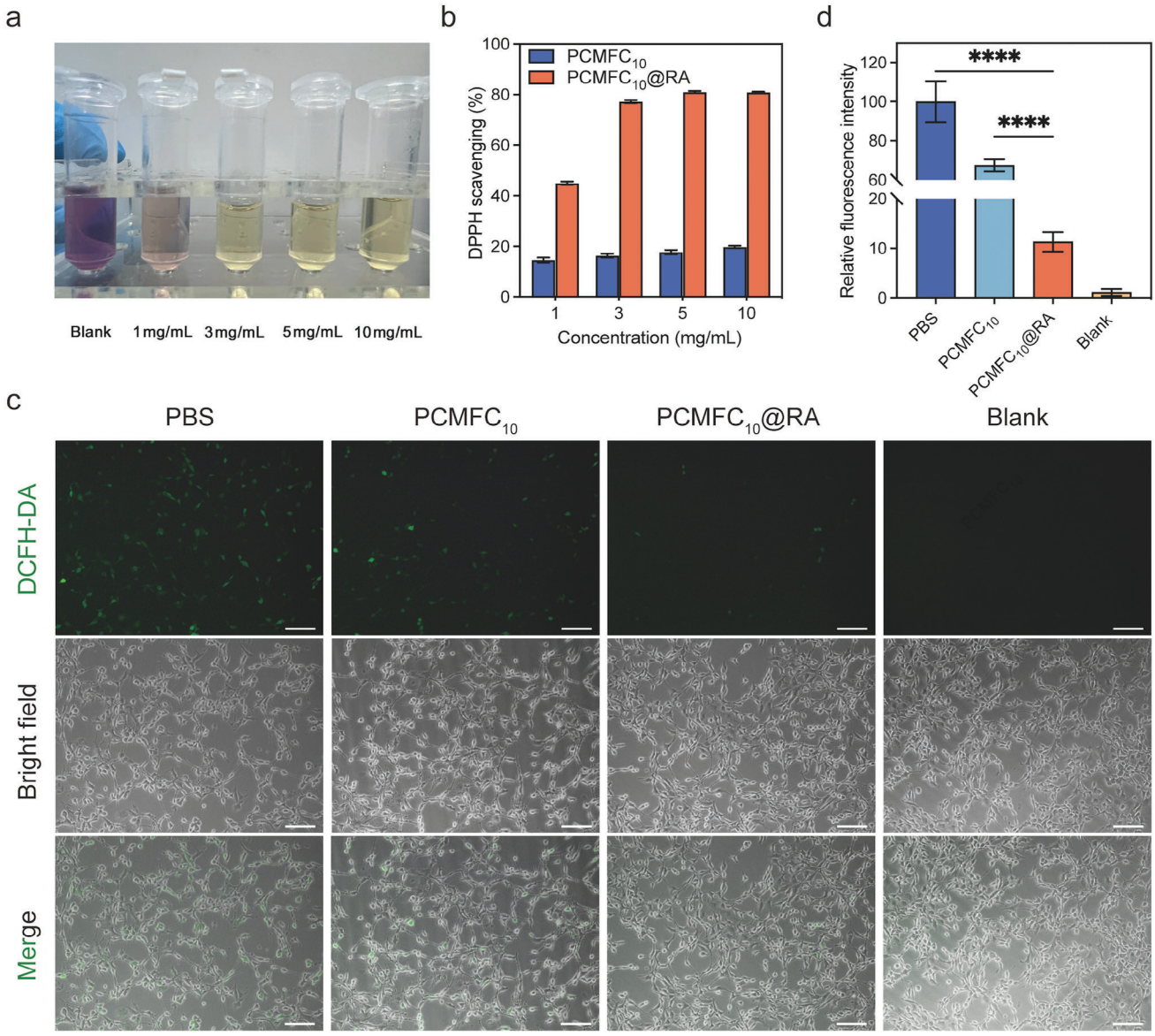


Figure 3. In vitro antioxidant capacity of PCMFC@RA hydrogels. a) Photograph of DPPH· scavenging test. b) DPPH· scavenging abilities of $PCMFC_{10}$ hydrogels and $PCMFC_{10}$@RA hydrogels at different concentrations. c,d) DCFH-DA fluorescence images and fluorescence intensity of NIH-3T3 cells after various treatments. Scale bar: 100 μm. All data are presented as mean ± SD ($n = 3$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

As a common oxidative stress damage agent, *tert*-butyl hydroperoxide (*t*-BHP), which is stable toward thermal decomposition, has basically the same damage mechanism as hydrogen peroxide.[33] To further verify the reactive oxygen species (ROS) scavenging ability of the hydrogels at the cellular level, NIH-3T3 cells were treated with *t*-BHP to simulate an oxidative stress environment, and the intracellular ROS levels were determined by DCFH-DA staining. According to Figure 3c,d, the fluorescence intensity of $PCMFC_{10}$ hydrogel group was not significantly reduced after treatment, implying its poor ROS scavenging ability. In contrast, the reduction of fluorescence intensity in $PCMFC_{10}$@RA hydrogel group indicated its excellent ROS scavenging ability. The above results demonstrated that rosmarinic acid endowed the hydrogel with significant antioxidant capacity, and $PCMFC_{10}$@RA hydrogel could effectively remove ROS generated in cells, thus relieving oxidative stress during the wound healing process.

2.2.4. In Vivo Hemostatic Performance

Film products such as Interceed have been approved by FDA for the prevention of postoperative adhesions, but effective hemostasis is required during their application, otherwise their barrier function will be compromised.[34] Therefore, materials that can safely and effectively prevent adhesion formation even with the

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

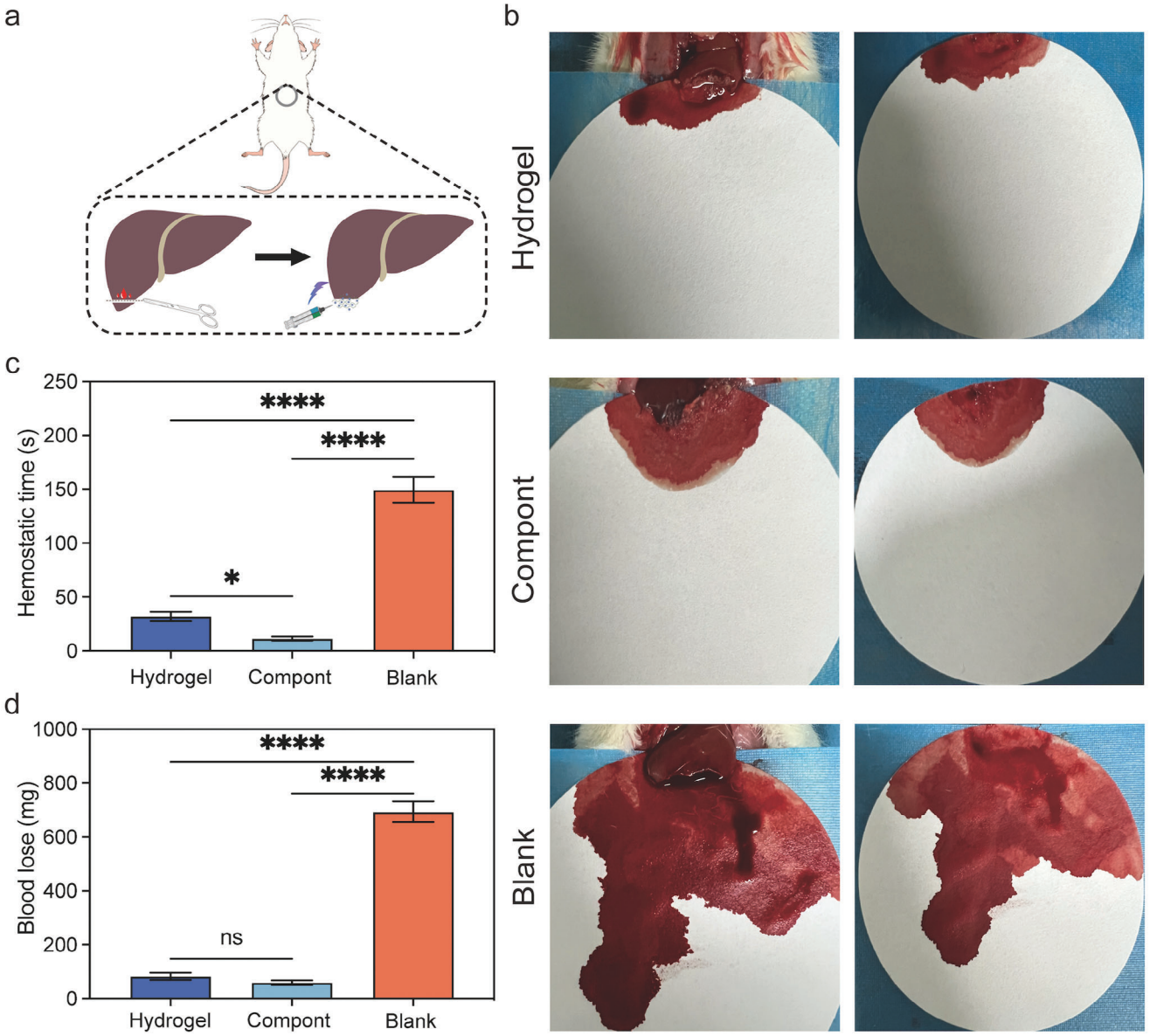


Figure 4. In vivo hemostatic performance of $PCMFC_{10}$@RA hydrogel. a) Schematic illustration of rat partial hepatectomy model. b–d) Bleeding conditions, hemostatic time, and blood loss of $PCMFC_{10}$@RA hydrogel, Compont medical glue, and blank control. All data are presented as mean ± SD ($n = 3$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

presence of blood or under conditions of incomplete hemostasis have more potential for clinical application.[35] The rat partial hepatectomy model was employed to evaluate the in vivo hemostatic ability of $PCMFC_{10}$@RA hydrogel (**Figure** 4a). Compont medical glue, which is commonly used in clinical practice, was used as control group, and no hemostatic treatment was used as blank group (Figure 4b). The hemostatic abilities of each group were evaluated by calculating the hemostasis time and blood loss. As shown in Figure 4c, compared with blank control group (149.5 s), the average bleeding time of hydrogel group was significantly reduced to 31.9 s, yet slightly more extended than Compont medical glue group (11.6 s). Quantitative statistical results (Figure 4d) revealed that the average blood loss in hydrogel group was 82.8 mg, which was slightly higher than that in Compont medical glue group (59.3 mg), but significantly lower than that in blank control group (693.6 mg). Although the hemostatic performance of $PCMFC_{10}$@RA hydrogel is slightly inferior to that of Compont medical glue with rapid curing ability, in actual use, Compont medical glue is very prone to misadhesion, and its hemostatic wound surface is relatively hard, which may cause secondary damage if not handled properly.[36] However, once the hydrogel prepolymerization solution is injected onto the incised surface, the Schiff base reaction occurs rapidly, and photocuring promotes hydrogel formation on the incised surface. The porous structure of the hydrogel is conducive to the rapid absorption of blood and acts as a physical barrier.[37] In addition, the catechol

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

groups in rosmarinic acid and the amino groups in chitosan can induce erythrocyte aggregation and platelet adhesion to accelerate the process of hemostasis.

2.3. Application of Hydrogel Polypropylene Composite Mesh in Abdominal Wall Defects Repair in Rats

2.3.1. Evaluation of Abdominal Adhesions and Defect Healing

Based on confirming the good anti-adhesion properties of hydrogels in vitro, next, we applied the $PCMFC_{10}$ hydrogel and $PCMFC_{10}$@RA hydrogel to the polypropylene (PP) mesh for the intraperitoneal repair of full-thickness abdominal wall defects in rats (Scheme 1b and Figure S6, Supporting Information). The rats were euthanized on day 7, 14, and 28 postoperation, and representative photos of abdominal adhesions were obtained (**Figure 5a**) and subsequently scored (Figure 5b–d). In all groups, the implants integrated well with the abdominal wall of the rats, and no mesh displacement was observed. In PP group, all rats showed abdominal viscera adhesion or omental adhesion, and had the highest adhesion score at each time point. The adhesion score in $PCMFC_{10}$-PP group was significantly lower than that in PP group, indicating that the hydration layer on the surface of hydrogel can alleviate the formation of adhesion to some extent. The lowest adhesion score was observed in $PCMFC_{10}$@RA-PP group, which can be attributed to the sustained release of rosmarinic acid in the hydrogel, helping to prevent adhesion formation further. In addition, over time, the adhesion scores in each group did not increase further, and even showed a slight downward trend. This phenomenon can be attributed to the fact that abdominal adhesion mainly occurs at 5–7 d after surgery, after which the new peritoneum gradually covers the surface of implant material to avoid the development of adhesion.[38,39] Taken together, our results illustrated that the $PCMFC_{10}$@RA hydrogel can effectively relieve abdominal adhesions caused by PP mesh implantation. The mechanism by which $PCMFC_{10}$@RA hydrogel coated on PP mesh reduces the formation of abdominal adhesions will be further elaborated in subsequent chapters from the perspectives of inflammatory response, oxidative stress, and fibrinolytic system activity.

2.3.2. Histological Analysis

Histological analysis was conducted using hematoxylin and eosin (H&E) and Masson's trichrome (MT) staining to evaluate the pathological changes in the healing process of abdominal wall defects. H&E staining results (**Figure 6**) showed that on day 7 after surgery, in PP group, the intestinal smooth muscle layer tightly adhered to the surface of PP mesh, and a large number of inflammatory cells infiltrated around the mesh, mainly neutrophils, suggesting that the adhesion formation process was closely related to inflammation. In $PCMFC_{10}$-PP group, abdominal adhesion occurred between omentum and mesh, and a certain degree of inflammatory cell infiltration was also observed due to acute foreign body reaction. In $PCMFC_{10}$@RA-PP group, only slight adhesion occurred, and the number of inflammatory cells around the mesh was significantly less than that in PP group and $PCMFC_{10}$-PP group. In addition, a few spindle-shaped mesothelial cells were seen above the inflammatory cell layer. On day 14, the inflammatory cell infiltration in PP group decreased, as fibroblasts, collagen fibers, and capillaries began to increase, and foreign body granulomas formed by the aggregation of multinucleated giant cells were observed. According to previous reports, the degradation of the hydrogels may contribute to the regeneration of granulation tissue, which is crucial for wound healing.[40,41] In $PCMFC_{10}$-PP group and $PCMFC_{10}$@RA-PP group, as hydrogels gradually degraded, the inflammatory cell layer was gradually replaced by tissue rich in collagen and fibroblasts, and complete peritoneal mesothelial cells could be observed covering the injury site, indicating the formation of new peritoneum. On day 28, the inflammation was further reduced in $PCMFC_{10}$-PP group, especially in $PCMFC_{10}$@RA-PP group, and only a few inflammatory cells were distributed around the mesh. However, there were still a certain number of neutrophils and multinucleated giant cells in PP group, and the inflammatory response was slightly weakened than before. Consistent with previous reports, inflammatory cells mainly accumulated on the surface of PP filaments, indicating chronic inflammation.[42,43] In addition, neovascularization around the mesh gradually increased over time in each group, which could provide oxygen and nutrients and promote defect healing.[44]

MT staining results (**Figure 7**) demonstrated minimal collagen deposition around the mesh in each group on day 7 due to the limited presence of fibroblasts during the acute inflammatory response. Over time, as the proliferation of fibroblasts, collagen deposition subsequently increased in each group. There were many dense collagen depositions at the adhesion site in the PP group, with a disordered arrangement, indicating the formation of permanent adhesion. The collagen deposition around the mesh in $PCMFC_{10}$-PP group and $PCMFC_{10}$@RA-PP group was significantly looser than that in PP group, which effectively alleviated the formation of adhesion. In addition, in each group, collagen deposition at the junction of the mesh and the abdominal wall was denser than that between the monofilaments of the mesh, further suggesting better integration of the mesh with the abdominal wall. It has been shown that severe inflammation is related to scar tissue density, and reduced inflammation mitigates fibrosis and promotes native tissue regeneration.[45,46] Based on our findings, we believe that $PCMFC_{10}$@RA-PP mesh can reduce the inflammatory response and inhibit excessive fibrosis, effectively promoting tissue regeneration. Coincidentally, it has been reported that F127-based hydrogels can effectively reduce the degree of endometrial fibrosis by downregulating TGF-β1 expression in uterine scar repair.[47]

2.3.3. Expression Levels of Pro-Inflammatory Cytokines

Inflammatory mediators play a prominent role in the formation of abdominal adhesions, among which tumor necrosis factor alpha (TNF-α) and interleukin 6 (IL-6) are considered pro-inflammatory cytokines associated with abdominal adhesions.[48,49] In order to further investigate the anti-inflammatory effect of the hydrogel, we assessed the tissue's TNF-α and IL-6 expression through reverse transcription quantitative real-time polymerase chain reaction (RT-qPCR) and

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

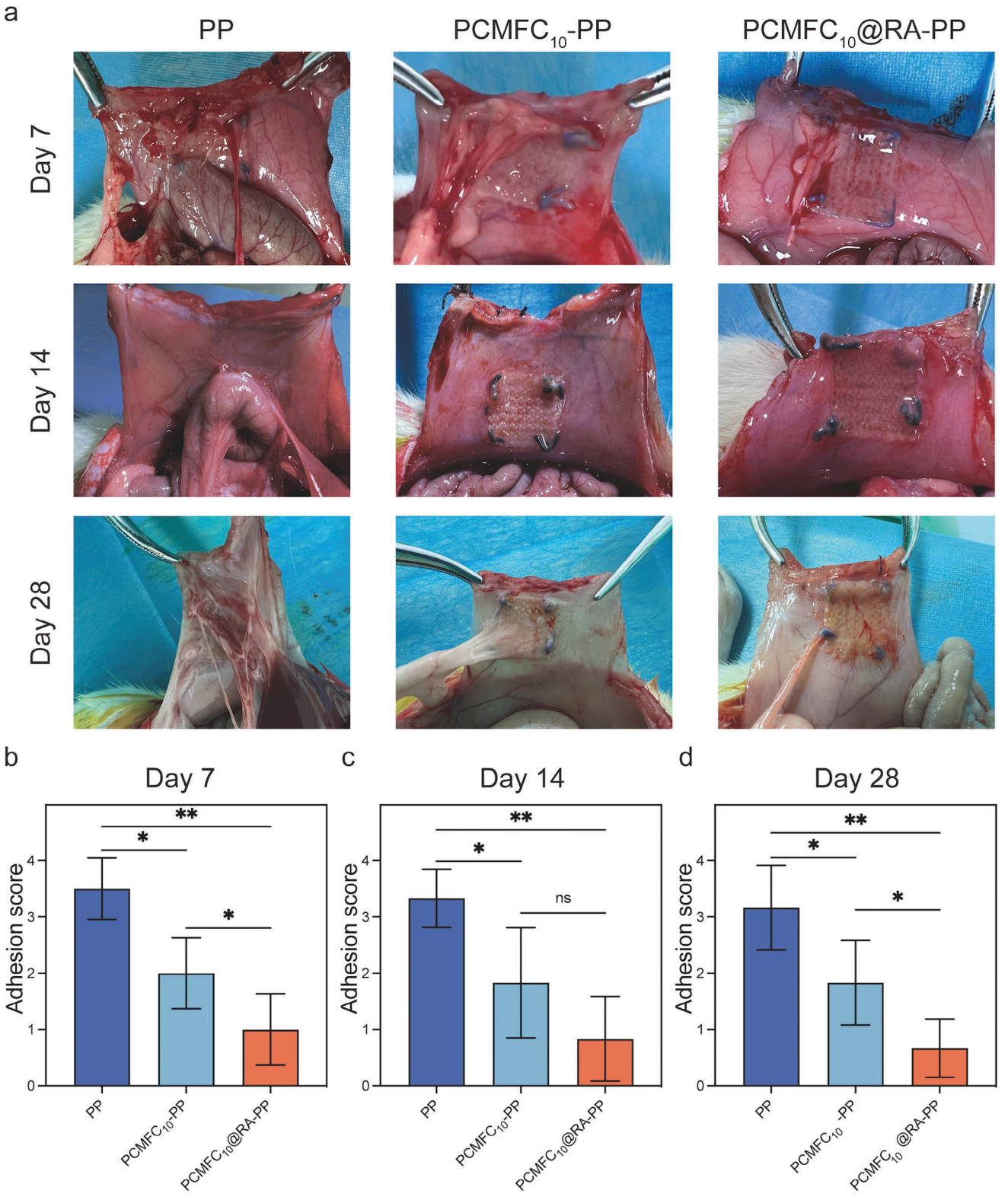


Figure 5. Evaluation of abdominal adhesions. a) Representative photos of abdominal adhesions on day 7, 14, and 28 postoperation. b–d) Adhesion scores on day 7, day 14, and day 28 postoperation. All data are presented as mean ± SD ($n = 6$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

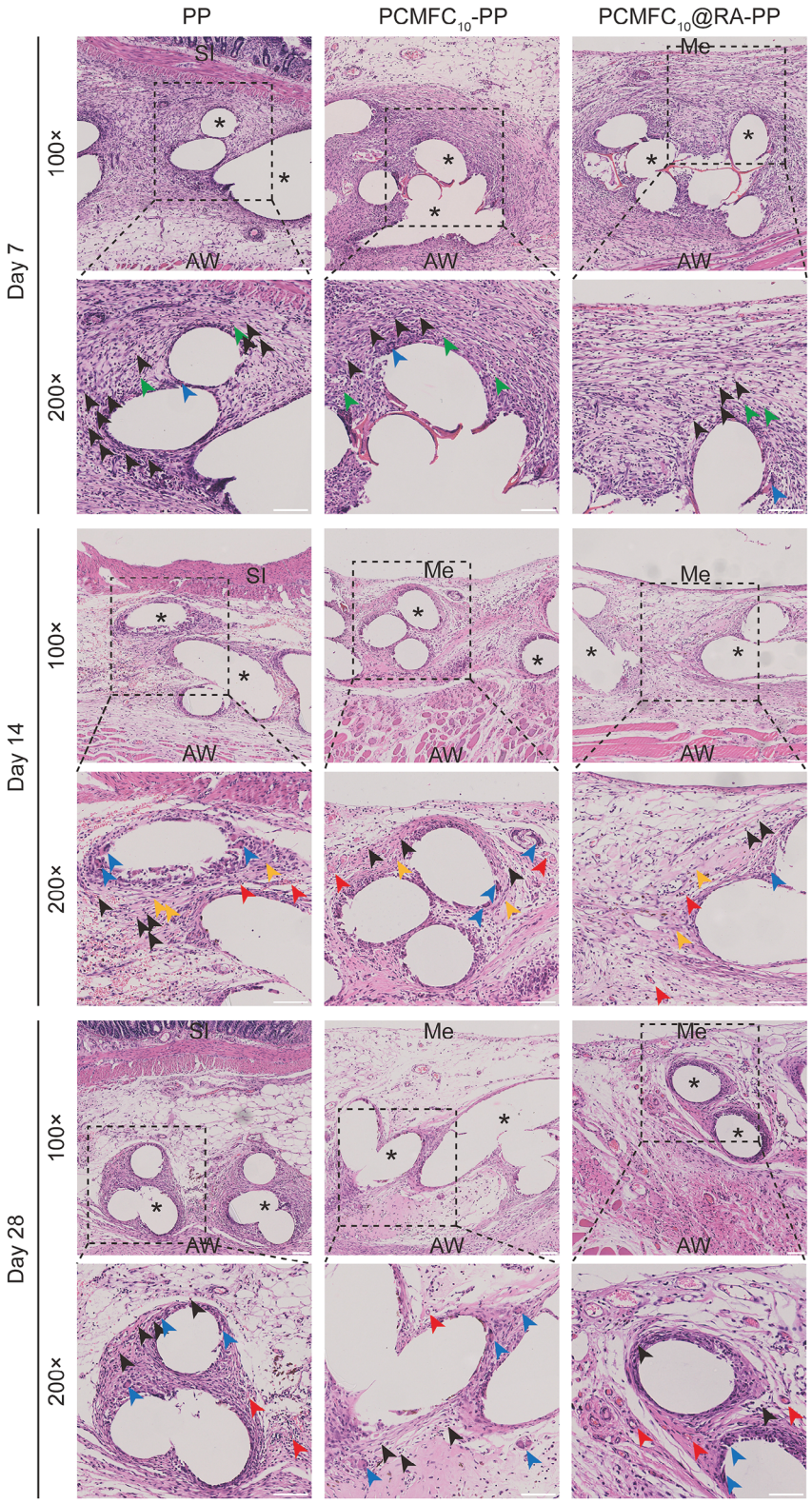


Figure 6. H&E staining of tissues in PP group, $PCMFC_{10}$-PP group, and $PCMFC_{10}$@RA-PP group at different time points. AW: abdominal wall; SI: small intestine; Me: mesothelial layer; "*": PP mesh; black arrows: neutrophils; green arrows: lymphocytes; blue arrows: multinucleated giant cells; yellow arrows: fibroblasts; red arrows: neovascularization. Scale bar: 100 μm.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

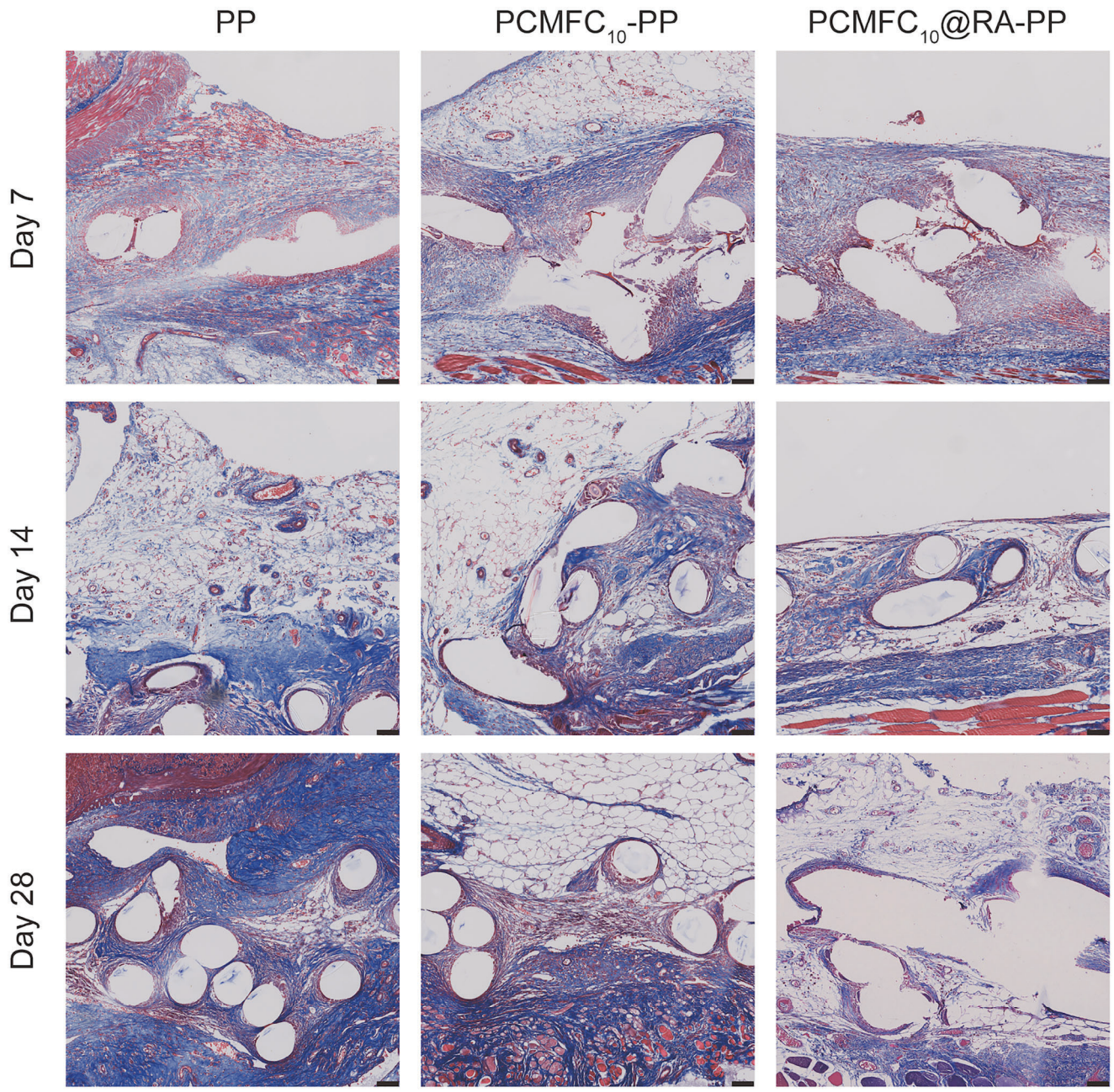


Figure 7. MT staining of tissues in PP group, $PCMFC_{10}$-PP group, and $PCMFC_{10}$@RA-PP group at different time points. Scale bar: 100 μm.

immunohistochemical (IHC) staining. RT-qPCR results showed that the relative mRNA expression levels of TNF-α (**Figure** 8a) and IL-6 (Figure 8b) in PP group were the highest at all time points. The level of TNF-α in $PCMFC_{10}$-PP group was not significantly different from that in the PP group, and the level of IL-6 was slightly lower than that in the PP group, indicating a limited inhibitory effect of $PCMFC_{10}$ hydrogel on TNF-α and IL-6. However, the mRNA levels of TNF-α and IL-6 in $PCMFC_{10}$@RA-PP group were significantly lower than those in the PP group.

We further detected the expression of TNF-α (**Figure** 9a) and IL-6 (Figure 9b) in the tissue around the mesh by IHC staining, and the results were consistent with those of RT-qPCR. On day 7, due to the acute inflammatory response, positive expression of TNF-α and IL-6 could be observed in all groups, and brown granules were found widely distributed around the mesh. As time went by, the positive expression of TNF-α and IL-6 around the mesh significantly decreased. On day 28, the lowest expression of inflammatory cytokines could be observed in $PCMFC_{10}$@RA-PP group, which was also reduced in $PCMFC_{10}$-PP group to some extent. In contrast, there were still higher levels of TNF-α and IL-6 expression around the mesh filaments in PP group, indicating the persistence of chronic inflammation. It can be seen that compared with PP mesh, $PCMFC_{10}$@RA-PP mesh has an excellent anti-inflammatory effect, which is not only due to the

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

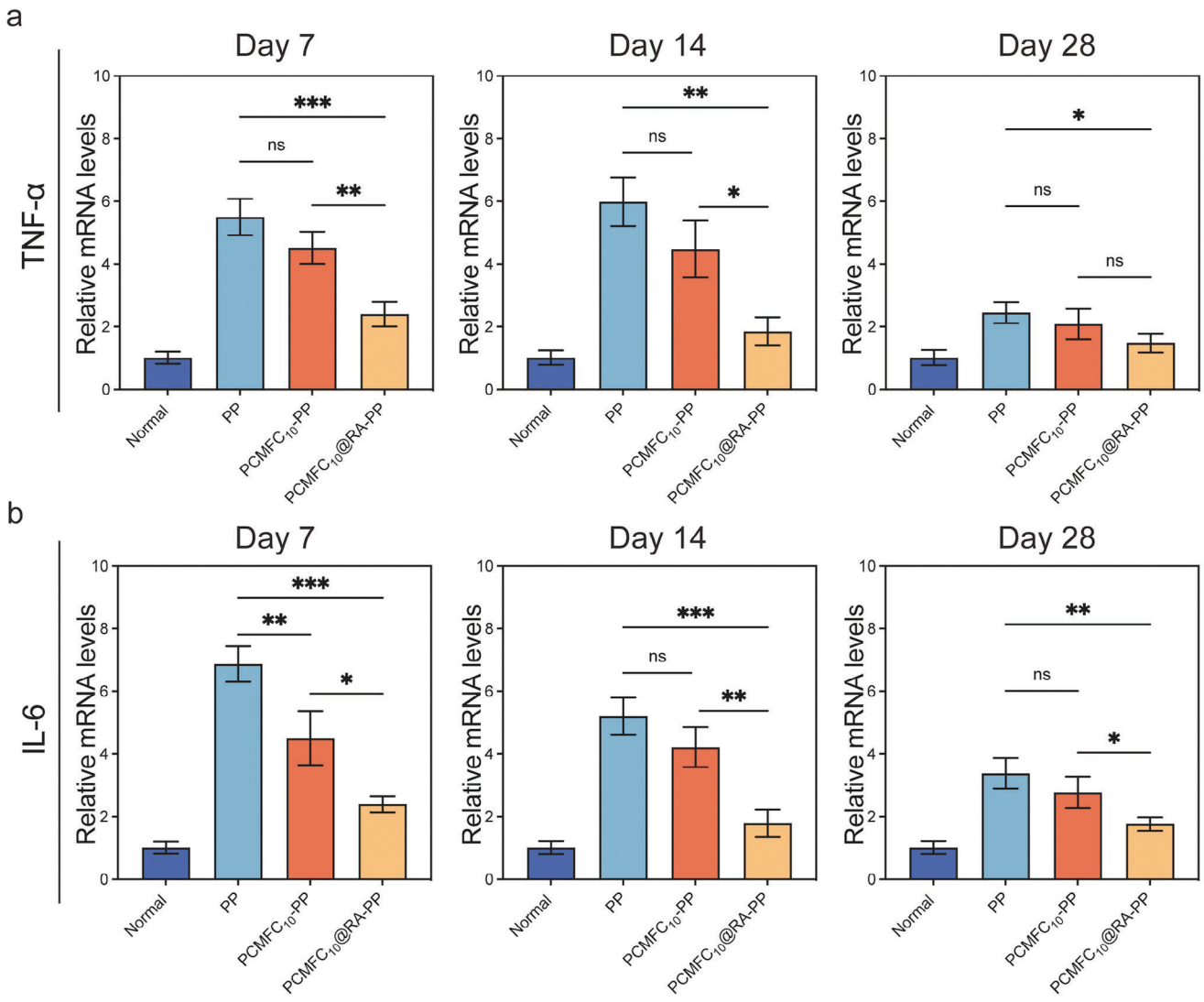


Figure 8. Pro-inflammatory cytokine levels detected by RT-qPCR. a,b) Relative mRNA expression levels of TNF-α and IL-6 in each group on day 7, 14, and 28 postoperation. All data are presented as mean ± SD ($n = 3$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

anti-inflammatory effects of chitosan in the hydrogel,[50] but also may be related to the powerful anti-inflammatory and antioxidant properties of rosmarinic acid.[51] In addition to this, the coating of hydrogel could prevent macrophages and inflammatory cells from infiltrating within the mesh, and the prevention of PP mesh from being exposed to the tissues may be the reason for reduced inflammation which resulted in the repair of hernia without adhesion.[34]

2.3.4. Oxidative Stress Levels

Inflammation and oxidative stress are inextricably linked to each other. Pro-inflammatory cytokines can cause oxidative stress by activating inflammatory cells, and in turn, oxidative stress can aggravate the inflammatory response[52] and further promote the formation of adhesions.[53] Previous studies have shown that the formation of postoperative adhesions may be avoided by scavenging ROS at the site of injury.[54,55] Rosmarinic acid, a natural phenolic acid compound, has attracted attention due to its excellent anti-inflammatory and antioxidant properties. Through in vitro experiments, we confirmed that $PCMFC_{10}$@RA hydrogel can effectively remove ROS produced in cells. Therefore, we hypothesized that the loading of rosmarinic acid in the hydrogel may alleviate oxidative stress in the repair of abdominal wall defects. To evaluate the degree of oxidative stress during the crucial period of postoperative adhesion formation, tissue samples collected on day 7 were detected for the level of lipid peroxide malondialdehyde (MDA) and the activities of endogenous antioxidant enzymes glutathione peroxidase (GPx) and superoxide dismutase (SOD). The results showed that the MDA levels of rats in both $PCMFC_{10}$-PP group and $PCMFC_{10}$@RA-PP group were significantly lower than those in the PP group, and the MDA level in $PCMFC_{10}$@RA-PP group was not significantly different

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

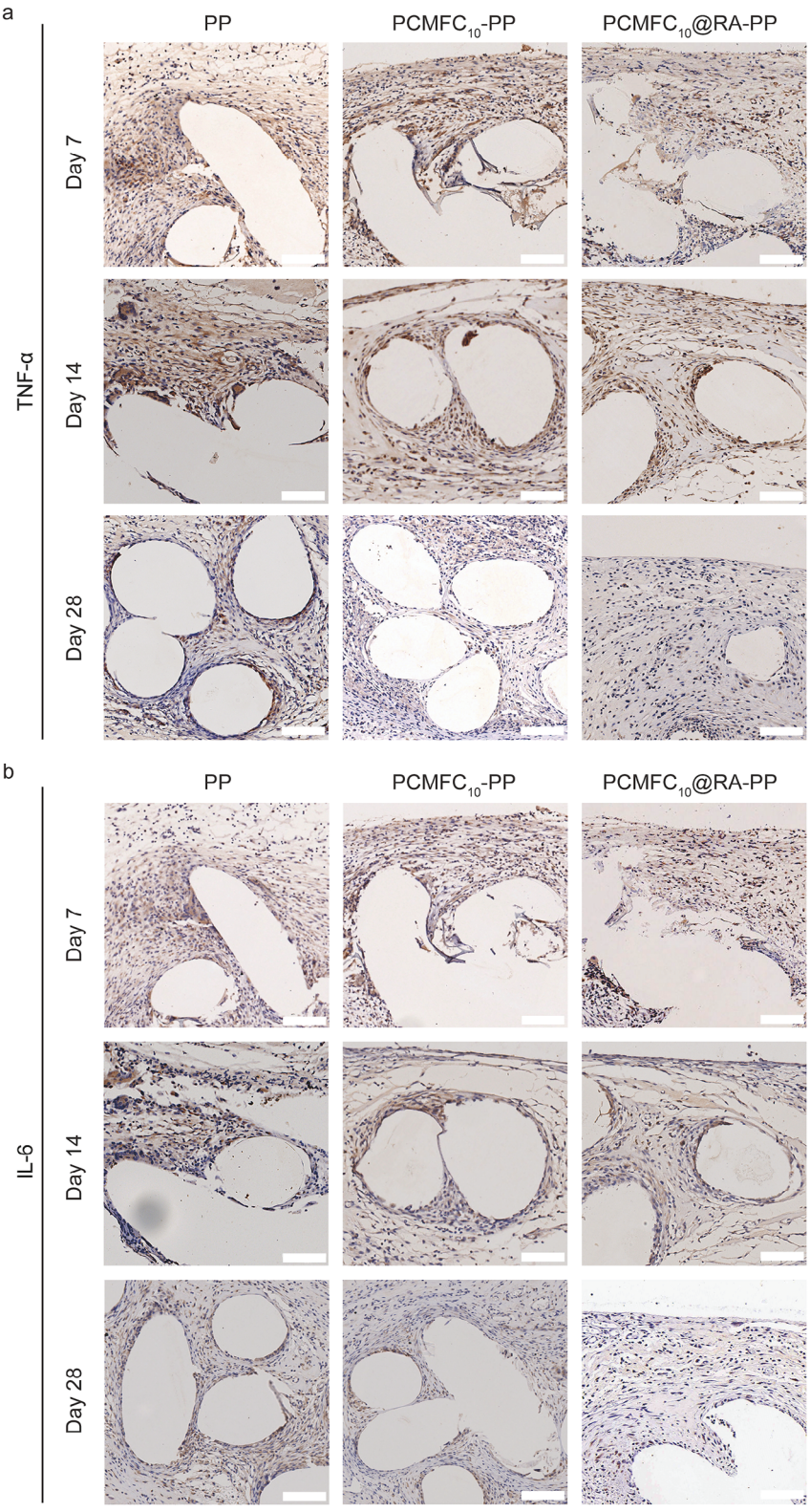


Figure 9. Pro-inflammatory cytokine levels detected by IHC staining. a,b) Images of TNF-α and IL-6 expression in each group on day 7, 14, and 28 postoperation. Scale bar: 100 μm.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

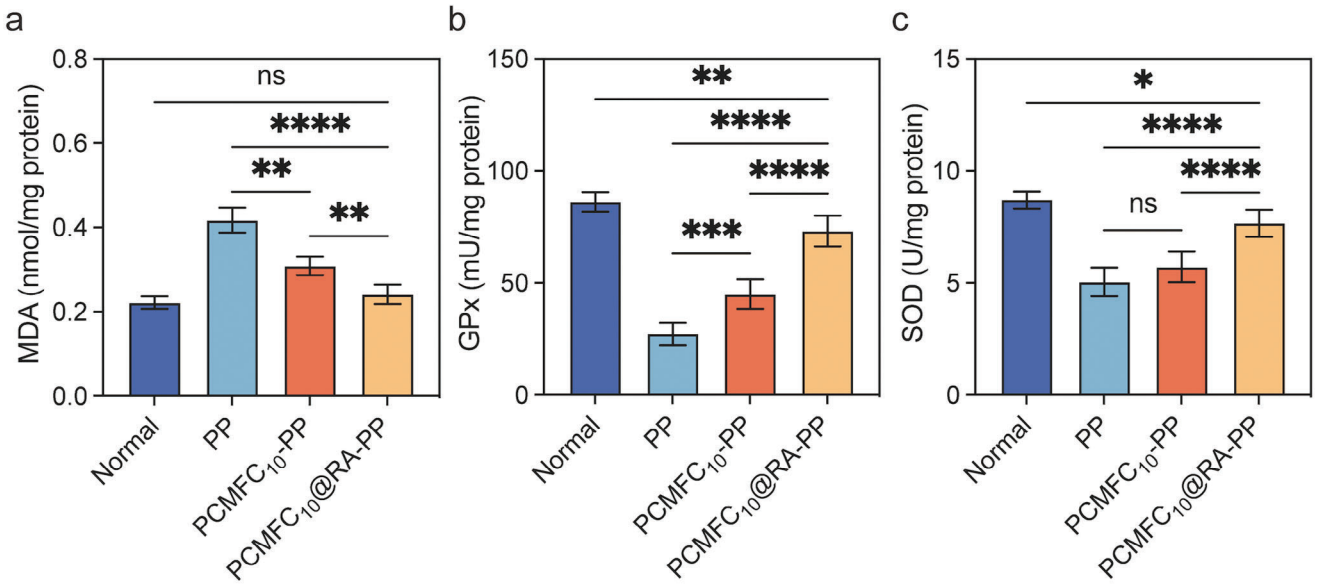


Figure 10. Oxidative stress levels. a–c) MDA level, GPx activity, and SOD activity in each group on day 7 postoperation. All data are presented as mean ± SD ($n = 6$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

from that in normal rats (**Figure 10a**). Correspondingly, PP group showed the lowest GPx activity, and after hydrogels treatment, the GPx activities were significantly improved (Figure 10b). Through the detection of SOD activity, it was found that the SOD activity in $PCMFC_{10}$@RA-PP group was significantly higher than that in PP group and $PCMFC_{10}$-PP group (Figure 10c). These results support the hypothesis that $PCMFC_{10}$@RA-PP mesh can reduce MDA level while increasing GPx activity and SOD activity, thereby alleviating abdominal adhesions by reducing oxidative stress. Consistent with the results of this study, previous study has shown that in rabbit abdominal wall defect models, higher levels of oxidative stress and inflammation can be observed with the implantation of uncoated PP mesh.[56] Two separate studies conducted in Iran have reported that *Rosemarinus officinalis* extract[57] and rosmarinic acid[58] could reduce abdominal adhesions while reducing pro-inflammatory cytokine levels and increasing antioxidant capacity.

2.3.5. Fibrinolytic System Activity

It is generally believed that the fibrinolytic system plays a crucial role in peritoneal healing. Tissue hypoxia at the surgical trauma site triggers oxidative stress and inflammatory responses, stimulating the exudation of fibrinogen-rich fluids. At the same time, the coagulation cascade reaction is activated, triggering the conversion of fibrinogen into fibrin. If fibrinolysis does not occur within 5–7 d after surgery, fibroblasts will produce collagen within the fibrin matrix, resulting in permanent adhesions. The fibrinolytic system is mainly activated by tissue plasminogen activator (t-PA) and inhibited by plasminogen activator inhibitor 1 (PAI-1). The imbalance of t-PA and PAI-1 can result in abnormal degradation and deposition of fibrin, both closely associated with the formation of postoperative adhesions.[59–61] Hence, the activity of the fibrinolytic system was investigated by RT-qPCR and IHC staining of t-PA and PAI-1 on day 7 postoperation. RT-qPCR results (**Figure 11a,b**) showed that PP group exhibited the lowest t-PA level and the highest PAI-1 level in comparison to the two hydrogel groups. Elevated t-PA levels were observed in $PCMFC_{10}$-PP group, especially in $PCMFC_{10}$@RA-PP group, while there was no significant difference in PAI-1 levels between the two hydrogel groups. This phenomenon indicates that the increased fibrinolytic activity dependent on rosmarinic acid is likely to be a result of increased t-PA activity. As shown in the double immunofluorescence staining images (Figure 11c), it can be intuitively found that PP group showed minimal red fluorescence (t-PA) and a large amount of green fluorescence (PAI-1). An increase in red fluorescence and a decrease in green fluorescence were detected in $PCMFC_{10}$-PP group, indicating that fibrinolytic activity could be improved to some extent, while the most red fluorescence and the least green fluorescence could be observed in $PCMFC_{10}$@RA-PP group. The above results indicated that after the implantation of conventional PP mesh, the overexpression of PAI-1 inhibited the activity of t-PA, and the fibrinolytic balance was disrupted, resulting in increased abdominal adhesion. On the contrary, the coating of hydrogel and the loading of rosmarinic acid in it can decrease the expression of PAI-1 while enhancing t-PA activity, accelerate the degradation of fibrin, and thus help to alleviate abdominal adhesions.

2.3.6. In Vivo Toxicity Evaluation

To evaluate the in vivo toxicity of the implants, all rats were observed and recorded after surgery. Throughout the experiment, all rats were in good general condition, with regular diet and excretion, and no unusual eye, nose, or mouth secretions were detected. The abdominal incisions of the rats healed normally, and no death, hernia formation, or other clinical complications occurred. To further assess the in vivo toxicity of the hydrogels,

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

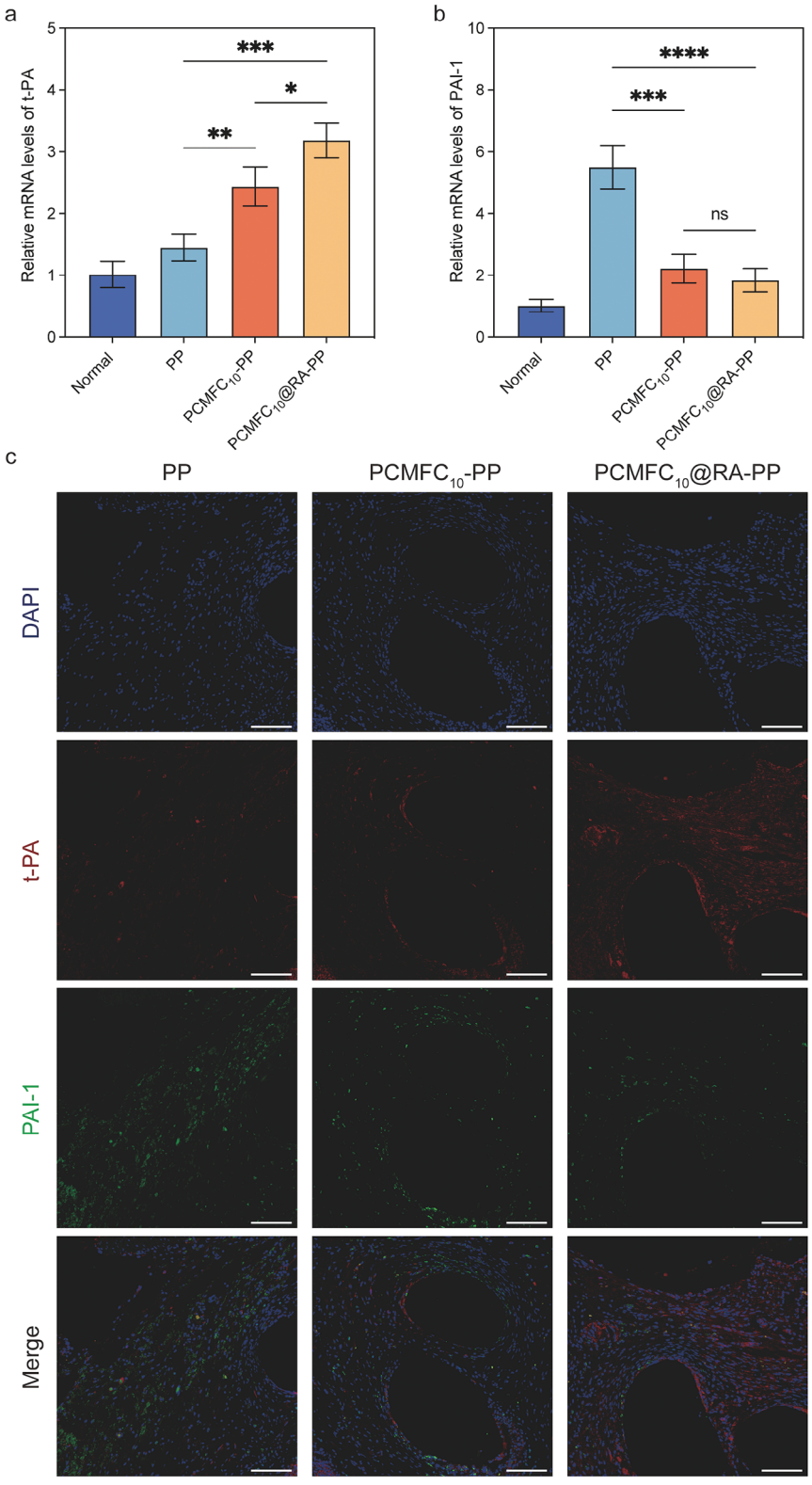


Figure 11. Fibrinolytic system activity. a,b) Relative mRNA expression levels of t-PA and PAI-1 in each group on day 7 postoperation. c) Double immunofluorescence staining images of t-PA and PAI-1 expression in each group on day 7 postoperation. Blue fluorescence: DAPI; red fluorescence: t-PA; green fluorescence: PAI-1. Scale bar: 100 μm. All data are presented as mean ± SD ($n = 3$ per group); the ns means no significant difference; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

major organs of rats in composite mesh groups were collected on day 7 after surgery. Gross observation revealed that the heart, liver, spleen, lungs, kidneys, and other major organs of rats in $PCMFC_{10}$-PP group and $PCMFC_{10}$@RA-PP group were basically the same as those of normal rats, with normal color and size, and no abnormal changes such as exudation, edema, and atrophy were visible to the naked eye. As shown in Figure S7 (Supporting Information), the above major organs were stained with H&E for further histopathologic examination. Compared to normal rats, the rats in two hydrogel-PP composite mesh groups exhibited no noticeable morphological changes toward the major organs, indicating that the two hydrogel-PP composite mesh had good in vivo biocompatibilities.

3. Conclusion

Direct contact between conventional PP mesh and abdominal organs may lead to abdominal adhesions and complications such as mesh erosion, intestinal obstruction, and intestinal fistulas, limiting its individual use in intraperitoneal repair. In this work, double crosslinked hydrogels with injectable and self-healing properties were prepared. We found that $PCMFC_{10}$@RA hydrogel had appropriate physicochemical properties and good in vitro biocompatibility. In the repair of full-thickness abdominal wall defects in rats, as a physical barrier, $PCMFC_{10}$@RA hydrogel could alleviate abdominal adhesions caused by conventional PP mesh implantation by reducing local inflammatory response, decreasing oxidative stress, and regulating the fibrinolytic system. Combined with the excellent tissue integration ability of PP mesh, $PCMFC_{10}$@RA-PP composite mesh is expected to become a new option for intraperitoneal repair. However, there are some limitations to this study. In order to better simulate human abdominal wall defects, it is necessary to further evaluate the role of $PCMFC_{10}$@RA-PP composite mesh in abdominal wall defect models of large animals, such as dogs and pigs. At the same time, the advantages and potential problems of the injectable hydrogel can also be explored through laparoscopic surgery. In this study, the PP mesh still needs to be sutured to the abdominal wall, and abdominal adhesions may occur at the knots. In subsequent studies, suture-free fixation mesh, such as biological glue and self-gripping mesh, can be used to reduce interference factors.

4. Experimental Section

The Experimental Section is available in the Supporting Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

Y.Z., X.L., and N.S. contributed equally to this work. This study was financially supported by the National Natural Science Foundation of China (No. 82300308), Shandong Provincial Natural Science Foundation (No. ZR2021MH119), and Shandong Provincial Natural Science Foundation (No. ZR2021MH158).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords

abdominal wall defects, abdominal adhesions, chitosan, hydrogel, polypropylenemesh, Pluronic F127, rosmarinic acid

Received: December 15, 2023
Revised: February 19, 2024
Published online: March 12, 2024

[1] J. W. Burger, R. W. Luijendijk, W. C. Hop, J. A. Halm, E. G. Verdaasdonk, J. Jeekel, *Ann. Surg.* **2004**, *240*, 578.
[2] R. Kaufmann, J. A. Halm, H. H. Eker, P. J. Klitsie, J. Nieuwenhuizen, D. van Geldere, M. P. Simons, E. van der Harst, M. van 't Riet, B. van der Holt, G. J. Kleinrensink, J. Jeekel, J. F. Lange, *Lancet* **2018**, *391*, 860.
[3] S. Todros, P. G. Pavan, A. N. Natali, *J. Biomed. Mater. Res., Part B* **2017**, *105*, 689.
[4] M. Kelly, K. Macdougall, O. Olabisi, N. McGuire, *Int. Urogynecol. J.* **2017**, *28*, 171.
[5] W. Hu, S. Lu, Y. Ma, P. Ren, X. Ma, N. Zhou, T. Zhang, Z. Ji, *J. Mater. Chem. B* **2017**, *5*, 575.
[6] D. Yang, Z. Song, J. Shen, H. Song, J. Yang, P. Zhang, Y. Gu, *Artif. Cells Nanomed. Biotechnol.* **2020**, *48*, 425.
[7] J. Zhu, R. E. Marchant, *Expert Rev. Med. Devices* **2011**, *8*, 607.
[8] P. Matricardi, C. Di Meo, T. Coviello, W. E. Hennink, F. Alhaique, *Adv. Drug Delivery Rev.* **2013**, *65*, 1172.
[9] M. H. Norahan, S. C. Pedroza-González, M. G. Sánchez-Salazar, M. M. Álvarez, G. Trujillo de Santiago, *Bioact. Mater.* **2023**, *24*, 197.
[10] W. Liang, W. He, R. Huang, Y. Tang, S. Li, B. Zheng, Y. Lin, Y. Lu, H. Wang, D. Wu, *Adv. Mater.* **2022**, *34*, 2108992.
[11] J. Gao, J. Wen, D. Hu, K. Liu, Y. Zhang, X. Zhao, K. Wang, *Bioact. Mater.* **2022**, *16*, 27.
[12] W. Hu, Z. Zhang, L. Zhu, Y. Wen, T. Zhang, P. Ren, F. Wang, Z. Ji, *ACS Biomater. Sci. Eng.* **2020**, *6*, 1735.
[13] K. H. Bae, L. S. Wang, M. Kurisawa, *J. Mater. Chem. B* **2013**, *1*, 5371.
[14] D. L. Taylor, M. in het Panhuis, *Adv. Mater.* **2016**, *28*, 9060.
[15] I. Y. Kim, S. J. Seo, H. S. Moon, M. K. Yoo, I. Y. Park, B. C. Kim, C. S. Cho, *Biotechnol. Adv.* **2008**, *26*, 1. http://doi.org/10.1016/j.biotechadv.2007.07.009
[16] M. Szulc, K. Lewandowska, *Molecules* **2022**, *28*, 247.
[17] N. M. Alves, J. F. Mano, *Int. J. Biol. Macromol.* **2008**, *43*, 401.
[18] A. Pitto-Barry, N. P. E. Barry, *Polym. Chem.* **2014**, *5*, 3291.
[19] A. V. Kabanov, E. V. Batrakova, V. Y. Alakhov, *J. Controlled Release* **2002**, *82*, 189.
[20] S. Noor, T. Mohammad, M. A. Rub, A. Raza, N. Azum, D. K. Yadav, M. I. Hassan, A. M. Asiri, *Arch. Pharmacal Res.* **2022**, *45*, 205.
[21] H. Guan, W. Luo, B. Bao, Y. Cao, F. Cheng, S. Yu, Q. Fan, L. Zhang, Q. Wu, M. Shan, *Molecules* **2022**, *27*, 3292.
[22] A. Wüst Zibetti, A. Aydi, C. A. Claumann, A. Eladeb, M. Adberraba, *J. Mol. Liq.* **2016**, *216*, 370.
[23] J. Wang, G. Li, T. Rui, A. Kang, G. Li, T. Fu, J. Li, L. Di, B. Cai, *RSC Adv.* **2017**, *7*, 9057.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[24] L. Song, L. Li, T. He, N. Wang, S. Yang, X. Yang, Y. Zeng, W. Zhang, L. Yang, Q. Wu, C. Gong, *Sci. Rep.* **2016**, *6*, 37600.
[25] G. M. Boland, R. J. Weigel, *J. Surg. Res.* **2006**, *132*, 3.
[26] C. Ding, L. Zhao, F. Liu, J. Cheng, J. Gu, S. Dan, C. Liu, X. Qu, Z. Yang, *Biomacromolecules* **2010**, *11*, 1043.
[27] S. Bazban-Shotorbani, M. M. Hasani-Sadrabadi, A. Karkhaneh, V. Serpooshan, K. I. Jacob, A. Moshaverinia, M. Mahmoudi, *J. Controlled Release* **2017**, *253*, 46.
[28] S. Chen, L. Li, C. Zhao, J. Zheng, *Polymer* **2010**, *51*, 5283.
[29] V. A. Feitosa, V. C. d. Almeida, B. Malheiros, R. D. d. Castro, L. R. S. Barbosa, N. N. P. Cerize, C. d. O. Rangel-Yagui, *Colloids Surf., B* **2019**, *180*, 177.
[30] ASTM International, Philadelphia, PA **2017**, https://doi.org/10.1520/F0756-17
[31] Y. Liang, X. Zhao, T. Hu, B. Chen, Z. Yin, P. X. Ma, B. Guo, *Small* **2019**, *15*, 1900046.
[32] S. Guo, Y. Ren, R. Chang, Y. He, D. Zhang, F. Guan, M. Yao, *ACS Appl. Mater. Interfaces* **2022**, *14*, 34455.
[33] C. F. Lima, M. Fernandes-Ferreira, C. Pereira-Wilson, *Life Sci.* **2006**, *79*, 2056.
[34] S. Konar, R. Guha, B. Kundu, S. Nandi, T. K. Ghosh, S. C. Kundu, A. Konar, S. Hazra, *Hernia* **2017**, *21*, 125.
[35] A. K. Singh Chandel, S. Ohta, M. Taniguchi, H. Yoshida, D. Tanaka, K. Omichi, A. Shimizu, M. Isaji, K. Hasegawa, T. Ito, *Biomater. Adv.* **2022**, *137*, 212825.
[36] X. Wei, C. Cui, C. Fan, T. Wu, Y. Li, X. Zhang, K. Wang, Y. Pang, P. Yao, J. Yang, *Int. J. Biol. Macromol.* **2022**, *199*, 401.
[37] D. Zhang, L. Mei, Y. Hao, B. Yi, J. Hu, D. Wang, Y. Zhao, Z. Wang, H. Huang, Y. Xu, X. Deng, C. Li, X. Li, Q. Zhou, Y. Lu, *Biomater. Res.* **2023**, *27*, 56.
[38] J. W. A. Burger, J. A. Halm, A. R. Wijsmuller, S. ten Raa, J. Jeekel, *Surg. Endosc.* **2006**, *20*, 1320.
[39] M. L. Baptista, M. E. Bonsack, J. P. Delaney, *Surgery* **2000**, *128*, 86.
[40] C.-Y. Zou, X.-X. Lei, J.-J. Hu, Y.-L. Jiang, Q.-J. Li, Y.-T. Song, Q.-Y. Zhang, J. Li-Ling, H.-Q. Xie, *Bioact. Mater.* **2022**, *16*, 388.
[41] Y. Zhao, B. Yi, J. Hu, D. Zhang, G. Li, Y. Lu, Q. Zhou, *Adv. Funct. Mater.* **2023**, *33*, 2300710.
[42] W. Hu, Z. Zhang, S. Lu, T. Zhang, N. Zhou, P. Ren, F. Wang, Y. Yang, Z. Ji, *Biomater. Sci.* **2018**, *6*, 3030.
[43] X. Yin, Y. Hao, Y. Lu, D. Zhang, Y. Zhao, L. Mei, K. Sui, Q. Zhou, J. Hu, *Adv. Funct. Mater.* **2021**, *31*, 2105614.
[44] J. M. Reinke, H. Sorg, *Eur. Surg. Res.* **2012**, *49*, 35.
[45] B. Klosterhalfen, U. Klinge, V. Schumpelick, *Biomaterials* **1998**, *19*, 2235.
[46] Y. Qiao, Q. Zhang, Q. Wang, Y. Li, L. Wang, *Acta Biomater.* **2021**, *128*, 277.
[47] Q. Hu, N. Xie, K. Liao, J. Huang, Q. Yang, Y. Zhou, Y. Liu, K. Deng, *Int. J. Biol. Macromol.* **2022**, *219*, 96.
[48] D. R. Ambler, N. M. Fletcher, M. P. Diamond, G. M. Saed, *Syst. Biol. Reprod. Med.* **2012**, *58*, 324.
[49] J. M. Anderson, A. K. McNally, *Semin. Immunopathol.* **2011**, *33*, 221.
[50] J. Qiao, Z. Jiang, Y. Yang, J. Chi, X. Qiao, B. Han, W. Liu, *Carbohydr. Polym.* **2017**, *172*, 255.
[51] A. Sadeghi, A. R. Bastin, H. Ghahremani, A. H. Doustimotlagh, *Mol. Biol. Rep.* **2020**, *47*, 3557.
[52] H. N. Siti, Y. Kamisah, J. Kamsiah, *Vasc. Pharmacol.* **2015**, *71*, 40.
[53] S. ten Raa, M. P. van den Tol, W. Sluiter, L. J. Hofland, C. H. J. van Eijck, H. Jeekel, *J. Surg. Res.* **2006**, *136*, 45.
[54] S. J. Heydrick, K. L. Reed, P. A. Cohen, C. B. Aarons, A. C. Gower, J. M. Becker, A. F. Stucchi, *J. Surg. Res.* **2007**, *143*, 311.
[55] Y. Wu, G. Wei, J. Yu, Z. Chen, Z. Xu, R. Shen, T. Liang, L. Zheng, K. Wang, X. Sun, X. Li, *Oxid. Med. Cell. Longevity* **2019**, *2019*, 4591384.
[56] V. Belebecha, R. Casagrande, M. R. Urbano, J. Crespigio, R. M. Martinez, D. L. Vale, S. H. M. de Almeida, *Int. Urogynecol. J.* **2020**, *31*, 139.
[57] Y. Roohbakhsh, V. B. Rahimi, S. Silakhori, H. Rajabi, P. Rahmanian-Devin, A. Samzadeh-Kermani, H. Rakhshandeh, M. Hasanpour, M. Iranshahi, S. H. Mousavi, V. R. Askari, *Planta Med.* **2020**, *86*, 405.
[58] A. Kakanezhadi, M. Rezaei, A. Raisi, O. Dezfoulian, F. Davoodi, H. Ahmadvand, *Sci. Rep.* **2022**, *12*, 18593.
[59] J. Tang, Z. Xiang, M. T. Bernards, S. Chen, *Acta Biomater.* **2020**, *116*, 84.
[60] B. Schnüriger, G. Barmparas, B. C. Branco, T. Lustenberger, K. Inaba, D. Demetriades, *Am. J. Surg.* **2011**, *201*, 111.
[61] B. W. J. Hellebrekers, T. Kooistra, *Br. J. Surg.* **2011**, *98*, 1503.

© 2024 Wiley-VCH GmbH

21922659, 2024, 15, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adhm.202304489 by Sichuan University, Wiley Online Library on [28/08/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License